

PRODUCT THINKING CASE STUDY · VEDANTU PM INTERNSHIP

Designing Consent & Data Trust for Young Viewers on YouTube

*Aligning YouTube's consent & data experience in India with the
Digital Personal Data Protection (DPDP) Act, 2023 - without breaking the product.*

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Minors are watching YouTube - but consent wasn't built for them

The Problem

- YouTube's sign-up flow relies on self-declared age with no real verification.
- Consent is one bundled “Agree & Continue” - it doesn't separate core service, personalisation and ad-data use.
- DPDP 2023 requires verifiable parental consent (VPC) and purpose-limited processing for anyone under 18.
- Scope: focusing on the account sign-up + Supervised Experience consent flow for under-18 users in India.

Why it matters

- Legal risk: penalties up to ₹250 crore for mishandling children's data.
- Trust risk: parents are the real decision-makers for these accounts.
- Reputational risk if flagged as a large-scale non-compliant platform.

Key Assumptions

- India-only scope; Google Family Link is today's baseline for Supervised Experience.
- No India-specific VPC flow exists today beyond generic parental setup.
- Mock data used throughout for metrics & dashboard.

Two people are affected - only one can legally consent



The Parent

Manages a supervised account for their 12-14 yr old

- Context: shares a family device; sets up Family Link once and rarely revisits it.
- Pain point: consent is a one-time generic toggle - no clarity on what data is collected or why.
- Pain point: no simple way to view, correct or delete their child's data later.
- Wants: control without friction - visibility, not more app-switching.

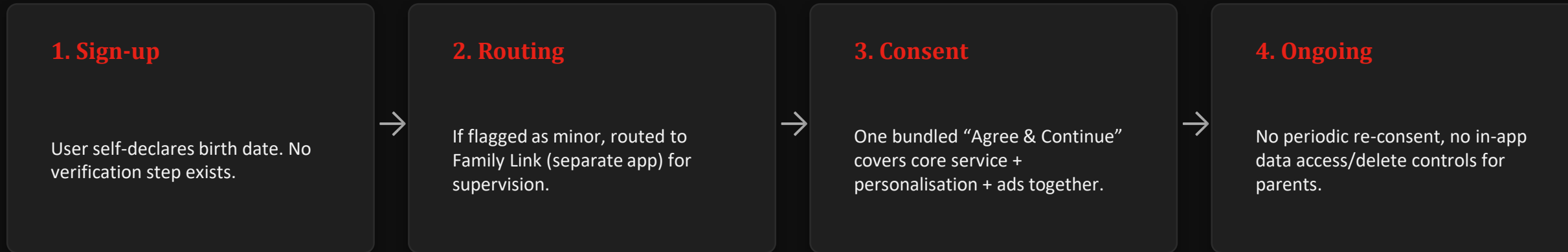


The Self-Signed-Up Teen

15-17 yr old who entered a false birth year at sign-up

- Context: wants an “adult” account to avoid restrictions and content filters.
- Pain point (systemic): never routed to any parental consent - fully invisible to DPDP protections.
- Risk: personal data (watch/search history, location) used for ad-targeting without any real consent.
- Why this matters: this is the highest-risk, highest-volume gap in the current system.

Where today's flow falls short of DPDP



Gaps vs. DPDP Expectations

- No verifiable parental consent (VPC) - self-declared age is gameable.
- Consent isn't purpose-specific or granular, as DPDP expects for children's data.
- No accessible data rights flow (access / correction / erasure) owned by the parent.
- No re-consent trigger when a supervised user ages into an adult account.

A layered consent system, built into surfaces users already use

1

Smarter Age-Assurance

Behavioural + account signals flag likely-minor accounts more reliably than self-declared birthdate alone - no mandatory ID upload for everyone.

2

Granular Consent

Four separate toggles for flagged minors: core service (required), personalised recommendations, ad personalisation, 3rd-party sharing - each explicitly opted in by the parent.

3

In-App Parent Dashboard

Data controls live inside YouTube's Supervised Experience settings, not a separate app: view data collected, withdraw consent, request access/delete in one tap.

4

Age-Up Re-consent

Fresh consent prompt triggered automatically when a supervised account turns 18, instead of silently carrying old settings forward.

One North Star, guarded on both sides

NORTH STAR METRIC

Verified Consent Rate

% of flagged-minor accounts with valid, current, purpose-specific parental consent

Parent Dashboard Engagement

% parents opening data controls monthly

7-Day Consent Completion

New flagged-minor accounts consented within 7 days

Rights-Request Resolution Time

Median hours to fulfil access / delete requests

Guardrail Metrics

- Minor sign-up completion / retention rate (must not crash)
- Session length on supervised accounts (experience must hold up)

Success Criteria (2 quarters post-launch)

- Verified Consent Rate $\geq 90\%$
- Rights requests resolved within 48 hrs for 95% of cases; sign-up completion drop $\leq 5\%$

North Star drops 15% in week 3 - no release shipped. Now what?

Hypothesis Tree

- A. Instrumentation bug - consent status stopped updating correctly in logs
- B. Funnel drop-off - a UX step is causing higher parent abandonment
- C. Denominator shift - new cohort (e.g. school term start) spikes flagged-minor sign-ups, diluting the rate
- D. External dependency change - a Family Link API/policy update broke the integration

What I'd Analyse

- Step-by-step funnel conversion, segmented by platform, OS version, region and signup cohort
- Denominator vs numerator trend separately - did the base of flagged minors jump?
- Recent deploy/experiment log and any Family Link API error rates

Root Cause & Next Steps

- Cross-tab the metric by day / platform / region to isolate where the drop concentrates
- Bug → hotfix; funnel drop → simplify the friction screen + parent reminder nudges; denominator shift → adjust metric window or let it normalise

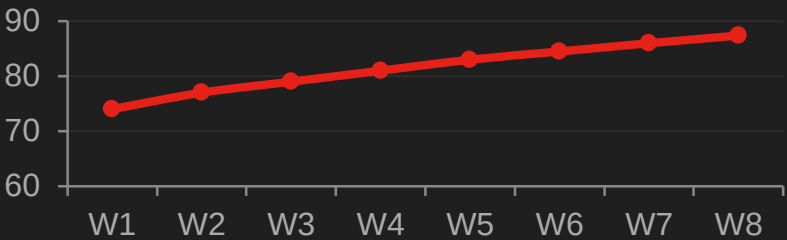
Monitoring consent health at a glance

MAIN KPI

87.4%

Verified Consent Rate ▲ 3.1 pts vs last month

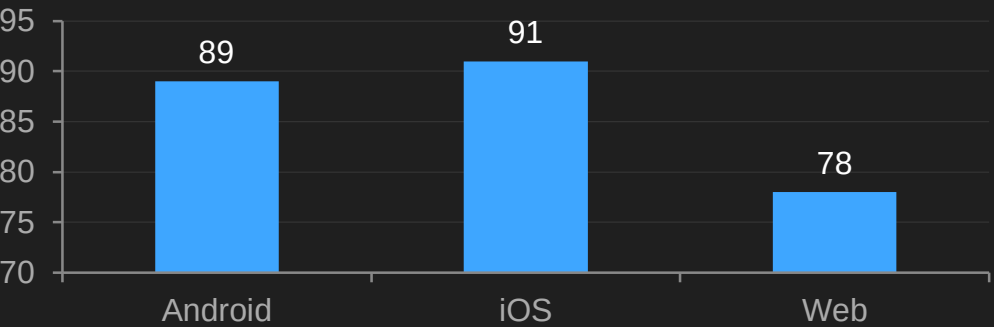
Trend (8 weeks)



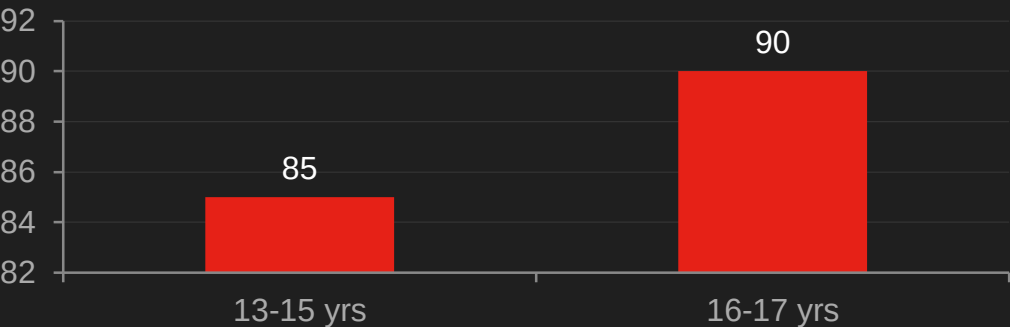
SUPPORTING METRICS

- Dashboard engagement: 34%
- 7-day consent completion: 91%
- Rights-request resolution: 29 hrs median

Segmentation: by Platform



Segmentation: by Age Band



Phased launch, with honest limits on A/B testing

Phase 1 - Soft Launch

5% of newly-flagged minor accounts in 2 states (Maharashtra, Karnataka). Watch funnel completion and early consent rate.

Phase 2 - Controlled Test

A/B test UX variants (toggle order, copy, nudge timing) on new sign-ups. NOT testing whether consent is asked - that's a legal baseline for everyone.

Phase 3 - Full Rollout

100% of India, dashboard live, quarterly consent-refresh reminders, existing accounts migrated in phased batches with monitoring.

Why compliance itself isn't A/B-testable

Withholding a legally required consent flow from a random holdout group isn't a valid control - it creates real compliance exposure for those users. So we test the how (UX copy, flow order, nudges) via A/B, and roll out the whether (must ask for consent) to 100% in phased, monitored batches instead.

Nothing here is free - here's what we're accepting

Engineering

Central consent-ledger must integrate YouTube app + Family Link + age-flagging logic - non-trivial build.

Business

Extra consent steps and reduced ad-personalisation on minor accounts likely dent short-term engagement & ad revenue.

User Experience

Parents face more steps upfront; risk of “consent fatigue” leading to blind “accept all” - must design against dark patterns.

Edge Cases

Users who decline consent still need core functionality (no personalisation); teens who lie about age; deletion requests after a parent loses account access; mid-session age-outs.

Trade-offs Accepted

- Chose proportional, soft age-assurance over strict mandatory ID verification - better UX at the cost of some residual gaming risk.
- Accepted upfront engineering cost and a modest, expected dip in minor-segment ad revenue in exchange for genuine DPDP compliance and parent trust.